

Detective Spoof: SPARC Voice Conversion Detection

DSP-Inspired Physics Features and Ensemble Machine Learning

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GitHub: <https://github.com/Ventacularity/Spoof-Detection>

1. Background and Related Work

Computer-generated voice conversion systems such as SPARC [1] can convincingly clone a speaker's voice, enabling impersonation scams, identity fraud, and phishing attacks. Previous anti-spoofing work includes MFCC/LFCC methods, waveform-based deep learning systems such as RawNet2, and commercial detectors such as Resemble AI DETECT-3B [4]. The ASVspoof initiative [2,3] established EER (equal error rate) as the standard evaluation metric, showing that unprotected ASV systems degrade from 1.63% to 23.83% EER under voice conversion attacks. Unlike these black-box generalist approaches, we target SPARC-specific artifacts using interpretable physics-based DSP features.

2. Methods

2.1 Spoofing System - SPARC

SPARC [1] decomposes speech into articulatory features, source features (such as pitch and loudness), and speaker embeddings before synthesis through a HiFi-GAN vocoder at 16 kHz.

2.2 Dataset

Six datasets drawn from the VCTK corpus and resampled to 16kHz to match SPARC's synthesis pipeline:

Set	Folder	Size	Label	Usage
A	source-vctk1k	1000	Real	Training
B	target-vctk1k	1000	Real	Target speakers (voice texture donors)
C	converted-vctk1k	1000	Spoof	Training (A→B via SPARC)
D	source-vctk500	500	Real	Out-of-distribution test
E	target-vctk500	500	Real	Target speakers (unseen)
F	converted-vctk500	500	Spoof	Out-of-distribution test (D→E via SPARC)

Models trained on A vs C (1000 real vs 1000 spoof, balanced). D and F held out entirely.

2.3 Preprocessing

STFT applied with 25ms frames, 10ms hop, Hamming window $w[n] = 0.54 - 0.46 \cdot \cos(2\pi n / (N-1))$ to suppress spectral leakage and produces complex spectrogram $S(f,t)$ used for all feature extraction.

2.4 Feature Extraction - 7 Physics Features

Each feature targets a specific HiFi-GAN vocoder limitation. All features are z-scored before classification:

$$z = (x - \text{mean}_{\text{train}}) / \text{std}_{\text{train}}$$

Feature	Formula	Physical Motivation	d(1k)	d(500)
Spectral Flux Std	$\text{std}(\sum_f \max(0, S(f,t) - S(f,t-1)))$	Vocoder upsampling cannot reproduce bursty consonant transitions of real speech	1.915	2.820
Modulation Energy	$\sum_f \{f \text{ in } [2, 10] \text{ Hz}\} \text{FFT}(H(y))^2$	Jaw rhythm (2-10 Hz) absent in math-generated amplitude variation	1.818	1.984
Spectral Flux Max	$\max(\sum_f \max(0, S(f,t) - S(f,t-1)))$	Explosive stops (bilabial 'p') smoothed out by convolutional upsampling	1.514	2.211
Glottal Std	$\text{std}(y - \text{LPC16}\{y\})$	Real glottal pulses are chaotic; SPARC uses smooth sinusoidal excitation	1.342	1.602
Aliasing Mean	$\text{mean}(\text{RMS}_{\{5-8\text{kHz}\}}(S(f,t)))$	HiFi-GAN upsampling artifacts in 5-8kHz band	1.214	1.141
Aliasing Std	$\text{std}(\text{RMS}_{\{5-8\text{kHz}\}}(S(f,t)))$	Real speech varies more in high-frequency content frame to frame	1.004	0.701
Glottal Kurtosis	$\text{kurt}(y - \text{LPC16}\{y\})$	DROPPED because d=0.481, only 331/1000 consistency (near random)	0.481	0.557

2.5 Classifiers and Ensemble

Logistic Regression (LR): Weighted sum of 4 z-scored features through sigmoid: $P(\text{spoo}) = 1/(1+e^{-\text{score}})$. Learned weights: Spectral Flux Std (-2.08), Modulation Energy (-1.94), Aliasing Mean (-0.76), Aliasing Std (+0.61). We set the threshold at 0.45, which is below 0.50 to prioritize spoo recall since missing a spoo is costlier than a false alarm. The model was trained with 4-fold cross-validation.

XGBoost: 400 sequential decision trees on all 7 z-scored features. Each tree asks binary threshold questions on features. Each successive tree corrects the previous tree's errors (gradient boosting). All leaf scores accumulate through sigmoid and capture nonlinear feature interactions that LR cannot represent with a single hyperplane.

Ensemble: OR: spoo if $P_{\text{LR}} \geq 0.45$ OR $P_{\text{XGB}} \geq 0.50$. AND: both must agree. AVG: $(P_{\text{LR}} + P_{\text{XGB}})/2 \geq 0.45$. AVG is the most robust out-of-distribution strategy.

3. Metrics

We determined classification performance using Accuracy (fraction of all clips correctly labeled), Precision (of all clips predicted as spoo, the fraction that are truly spoo), Recall (of all actual spoofs, the fraction our model catches), and F1-Score (harmonic mean of precision and recall). For feature selection we use Cohen's d = $(\text{mean}_{\text{real}} - \text{mean}_{\text{spoo}}) / \text{pooled_std}$, which measures how many standard deviations apart the real and spoo distributions are, where values above 0.5 meant a medium effect and values above 1.0 indicate a large effect. We also report pair wise consistency, the fraction of matched real/spoo pairs where the real clip scores higher than the spoo clip on a given feature. Per class accuracy is reported alongside raw counts such as 990/1000 real correct and 992/1000 spoo correct to clearly distinguish performance on each class.

4. Results

4.1 Feature Discriminability

Cohen's d values exceed 1.0 for all 6 kept features on both datasets demonstrating a large effect by convention. Spectral Flux Std is strongest at d=1.915 (VCTK) and d=2.820 (VC500), with 943/1000 and 499/500 pair-wise consistency respectively. Cohen's d values are higher on the out-of-distribution VC 500 dataset than on the training-matched VCTK data, confirming these features detect genuine SPARC synthesis artifacts rather than memorizing training speakers. Glottal Kurtosis scored d=0.481 with only 331/1000 consistency (33.1%) and was dropped.

4.2 Classification Results

Strategy	Real Correct	Real %	Spoof Correct	Spoof %	Overall
— VCTK 1000-Pair (In-Distribution) —					
LR alone	885/1000	88.5%	970/1000	97.0%	92.8%
XGB alone	990/1000	99.0%	992/1000	99.2%	99.1%
OR	885/1000	88.5%	998/1000	99.8%	94.2%
AND	990/1000	99.0%	964/1000	96.4%	97.7%
AVG	929/1000	92.9%	996/1000	99.6%	96.2%
— VC 500-Pair (Out-of-Distribution) —					
LR alone	443/500	88.6%	499/500	99.8%	94.2%
XGB alone	452/500	90.4%	455/500	91.0%	90.7%
OR	435/500	87.0%	500/500	100.0%	93.5%
AND	460/500	92.0%	454/500	90.8%	91.4%
AVG	449/500	89.8%	498/500	99.6%	94.7%

XGBoost achieves 99.1% accuracy in-distribution with near-perfect probability separation, where real clips average a spoof probability of 0.027 and spoof clips average 0.981. On out-of-distribution data, XGBoost drops to 90.7% while Logistic Regression maintains 94.2% and the AVG ensemble achieves 94.7%. XGBoost's complex nonlinear boundaries overfit to training speaker characteristics while the linear hyperplane of Logistic Regression generalizes more reliably, a classic bias-variance tradeoff. The AVG ensemble benefits from their complementary strengths: Logistic Regression excels at spoof recall at 99.8% on VC 500 while XGBoost excels at real accuracy at 90.4%, and averaging the two yields the most robust out-of-distribution performance with only a 1.5% drop from in-distribution results.

5. Discussion and Failure Cases

5.1 Phase Variance Hypothesis - Null Result

Our original proposal hypothesized that real voices would exhibit higher phase variance than synthetic voices. Averaged analysis across 1000 pairs showed real mean phase variance = 3.271 and spoof mean = 3.257 which is a difference of 0.014 with Cohen's $d \sim 0$. SPARC's HiFi-GAN vocoder does not introduce phase regularization artifacts. This null result redirected our feature engineering toward spectral energy and temporal modulation features.

5.2 Class Imbalance

Initial training included target-vctk1k alongside source-vctk1k in the real class (2000 real vs 1000 spoof). This imbalance caused LR accuracy to drop to 67.9% with poor spoof recall. Removing target-vctk1k and training on a balanced 1000 vs 1000 restored LR to 92.8% and XGBoost to 99.1%. Balanced training is critical when the two error types carry different costs.

6. Conclusion and Future Work

Detective Spoof achieves 99.1% in-distribution (XGBoost) and 94.7% out-of-distribution (AVG ensemble) using 7 physics-based DSP features derived from the HiFi-GAN vocoder limitations. Spectral Flux Std ($d=1.915$) and Modulation Energy ($d=1.818$) are the strongest discriminators, capturing the vocoder's inability to reproduce natural speech burstiness and biological jaw rhythm. Unlike black-box approaches, every prediction is explainable in physical terms.

Future work include adding more target speakers in training to decouple speaker style from synthesis artifacts; test against other synthesis systems (ElevenLabs, Tortoise TTS); explore real-time detection; compute EER and AUC-ROC to

align with the ASVspoof community's evaluation standards.

7. Contributions

Erick Lara led research and conceptual design for identifying the 7 physics-based features, establishing feature selection criteria, researching existing anti-spoofing methods, and contributing to code development.

Ching Yuen and Ved Sheth implemented the full codebase which consisted of feature extraction, model training, ensemble evaluation, and all analysis and reporting scripts.

References

- [1] SPARC — Lian et al. 2024 <https://arxiv.org/abs/2406.12998>
- [2] ASVspoof 2019 Database — Wang et al. 2020 <https://doi.org/10.1016/j.csl.2020.101114>
- [3] ASVspoof 2019 Countermeasures — Nautsch et al. 2021 <https://ieeexplore.ieee.org/document/9358099>
- [4] Resemble AI DETECT-3B <https://www.resemble.ai/detect/>